

Impact of Milky Way Dust Extinction Maps on Supernova Ia model Training, Fitting, and Cosmology; Forecasts for LSST

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ABSTRACT

Accurate measurements of cosmic expansion with Type Ia supernovae rely on accurate independent measurements of interstellar dust absorption and reddening $E(B - V)$ in the Milky Way. We assess the impact of $E(B - V)$ maps from different background sources on precision cosmology using simulated supernova samples representative of the Vera C. Rubin Observatory's Legacy Survey of Space and Time (LSST). Thus far, SN cosmology analyses have applied Milky Way corrections at the light-curve fitting level; here we develop an improved pipeline to incorporate a selection of multiple dust maps into the Spectral Adaptive Light-curve Template (SALT3) model retraining. We also develop an approach to use the low-redshift supernova in the linear regime of the Hubble diagram to distinguish between maps, and show that future SN surveys such as LSST may have the statistical power and precision to rule out extinction maps. Importantly for the impact on cosmology, we show that the standard SN color correction in light curve fitting can fully compensate for color differences among MW dust maps, with differences in SN distances arising only from differences between the SN and MW color laws and coefficients, reaching ~ 0.025 mag at $E(B - V) = 0.05$. While previous SN cosmology analyses have accounted for MW dust uncertainties only during light curve fitting (with a 5% $E(B - V)$ scaling), we find that consistently propagating these uncertainties through SALT3 training substantially raises their cosmological impact, increasing the degradation in the year one LSST dark energy Figure of Merit by approximately 18%. This suggests that MW dust uncertainties for Stage-4 SN cosmology are among the most important systematic effects, on the level of calibration and extragalactic astrophysics. By providing a new framework for evaluating Galactic dust maps, this work lays the foundation for reducing Milky Way extinction systematics in future supernova cosmology analyses.

Keywords: Cosmology, cosmology: observations, (stars:) supernovae: general

1. INTRODUCTION

Since the discovery of the accelerating expansion of the Universe (Riess et al. 1998; Perlmutter et al. 1999), Type Ia Supernovae (SNe Ia) have become an essential tool for cosmology. Their use as “standardizable” candles allows their observed brightness to be an indicator of their distance, a strategy that has proven effective well beyond redshifts of $z = 1$. Over the past two decades,

surveys such as PanSTARRS (PS1; Jones et al. 2018) and the Dark Energy Survey (DES; Sánchez et al. 2024) have expanded the number of SNe Ia we use for cosmology, and compilations such as Pantheon+ (Brout et al. 2022), Union3 (Rubin et al. 2025), and DES-DOVEKIE (Popovic et al. 2026), have enabled precise constraints on previous cosmological analyses.

The next generation of all-sky telescope surveys promises to improve these samples by more than a factor

of 10 in the next decade. The Vera C. Rubin Observatory’s Legacy Survey of Space and Time (LSST) anticipates the discovery of over 100,000 SNe Ia with two survey strategies: Wide Fast Deep (WFD) and Deep Drilling Field (DDF) (Rosselli et al. 2025a). As a result, it has become important to quantify the anticipated statistical uncertainties, study the possible impact of systematic uncertainties on inferred cosmological parameters (The LSST Dark Energy Science Collaboration et al. 2018a), and leverage the benefits of large sample sizes while understanding the accompanying limitations.

For all-sky cosmology, one of the most significant sources of systematic uncertainty arises from interstellar dust within our own galaxy. Dust both extinguishes and reddens the light from distant sources, altering their observed brightness and colors and therefore biasing the inferred SN Ia distances (Brout & Riess 2023). To mitigate the effect of Milky Way (MW) dust, recent surveys including DES (Smith et al. 2020) focused on very low-extinction regions of the sky, with a mean MW color-excess of $E(B - V) < 0.07$. Color-excess being the difference between the observed and intrinsic source color in two given bands (usually B and V): $E(B - V) = (B - V)_{\text{observed}} - (B - V)_0$ (Brout & Riess 2023). Although this precaution reduces the impact of extinction, it relies on accurate mapping of MW dust and heavily limits observations to a small fraction of the sky. Surveys such as LSST will require more thoughtful consideration of the impact of MW dust on their findings (Ivezic et al. 2019), as many of these newly discovered SNe Ia will exist in regions of the sky with far more complicated MW extinction. This will be especially important for the acquisition of low- z samples that are volume-limited within $z < 0.1$ and that will rely solely on the WFD survey strategy. Understanding the impact dust uncertainties have on this population is especially important due to the critical role these nearby SNe play as cosmological model anchors in the Hubble Diagram (Tang et al. 2025), as a probe of the peculiar velocity field (Carreres et al. 2025), and as distance calibrators in the distance ladder (H0DN Collaboration et al. 2026).

Most modern supernova cosmology analyses have relied on the Schlafly et al. (2010) dust map, which recalibrated the original Schlegel et al. (1998) (SFD) map by applying an ~ 0.86 normalization factor, to correct for MW extinction during both spectrophotometric model training and light-curve fitting. These analyses have additionally adopted a 5% global scaling uncertainty on MW $E(B - V)$. This systematic is motivated by the approximately 4% difference in reddening normalization found by Schlafly & Finkbeiner (2011) (hereafter S11) relative to Schlafly et al. (2010), despite

the analyses probing a very similar Sloan Digital Sky Survey (SDSS) footprint. The conventional treatment of this MW extinction uncertainty in SN Ia cosmology has therefore been to propagate a global scaling of the assumed dust correction through the light-curve fitting stage (e.g. Jones et al. 2018; Brout et al. 2022; Vincenzi et al. 2024). DES reports a MW dust systematic that is directly comparable in size to many astrophysical systematics, such as those associated with SN Ia host galaxy dust and larger than the uncertainty from photometric classification (Vincenzi et al. 2024). Brout et al. (2022) identifies MW dust as one of the top three dominant sources of systematic uncertainty in the Pantheon+ error budget.

Recently, Karim et al. (2025) studied the impact of implementing alternative Galactic extinction maps in a galaxy clustering study and found over 1σ shift in $\sigma_8 - \Omega_M$. This work attempts to perform a comprehensive study of the uncertainty associated with MW dust corrections for SN Ia cosmology. We employ two key improvements over previous analyses: first, we examine the diversity among modern extinction maps in the literature and provide a more realistic estimate of this systematic, and second, we investigate the impact of implementing dust map systematics in the retraining of the SALT3 model by changing the extinction assumptions in training data. We focus exclusively on extinction map systematics in this study, leaving the impact of color laws to planned future work.

In recent years, many new MW dust maps have become available, many of which have been constructed using independent observational tracers. Improvements in sky coverage, angular resolution, and the availability of new datasets have enabled increasingly detailed estimates of Galactic extinction. Recent maps incorporating data from surveys such as the Dark Energy Spectroscopic Instrument (DESI) further expand the set of available extinction estimates (Zhou et al. 2025). These are just the latest addition to a growing collection of high-resolution extinction maps now being used in cosmological analyses (Chiang 2023; Planck Collaboration et al. 2014; Mudur et al. 2022; Cheng et al. 2025; Lenz et al. 2017). As the number and diversity of these maps continue to grow, it is increasingly important to systematically evaluate how the choice of MW dust map propagates into cosmological measurements.

In this work, we propagate differences among modern MW dust maps consistently through the SALT3 training, light-curve fitting, and cosmological inference for realistic LSST simulations. Rather than seeking to identify a uniquely preferred extinction map, we treat the diversity among current reconstructions as an empiri-

cal measure of uncertainty and quantify how that uncertainty propagates into SN Ia distances. This paper is organized as follows; §2 provides background information about the simulation framework, models, and dust maps used in this study; §3 details the methods used, including the dust map interpolation process and the use of these maps in training SALT models; and §4 presents our simulation results and map discrimination technique. In §5, we summarize our work.

2. BACKGROUND

2.1. LSST Simulations

To investigate the impact of MW dust systematics on the large SN Ia samples expected from the full ten-year Legacy Survey of Space and Time (LSST), we generate simulations using the Supernova Analysis package (SNANA; Kessler et al. 2009) and the helper pipeline Pippin (Hinton & Brout 2020). We use cadence files developed and tested for the Photometric LSST Astronomical Time-Series Classification Challenge (PLAsTiCC; Kessler et al. 2019). These cadence files encode the filter sequence, sky noise, seeing, and $E(B - V)$. We specifically choose the PLAsTiCC simulation parameters to avoid any impact that the LSST rolling observation cadence could have on our cosmological results.

The simulations are divided into a low-redshift ($0.02 < z < 0.17$) WFD sample and a high-redshift ($0.17 < z < 1.25$) DDF sample. The size of the WFD sample is tuned to match the ten year forecast in Rosselli et al. (2025b). We assume a spectroscopically classified sample and therefore do not include uncertainties associated with photometric classification. Following all analysis quality and common population cuts, our final, cosmology-grade sample includes 11,212 SNe Ia for WFD and 102,949 SNe Ia for DDF. Both simulations represent the full ten-year survey, use a pre-rolling cadence, and consist of observations in the LSST *griz* bands. We choose to exclude the *u* and *Y* bands from this work to simplify the process, a choice we are confident does not significantly effect our results as it does not influence the SALT3 training which we find to have the greatest impact.

This simulation framework provides a controlled environment in which the adopted MW dust map is the primary systematic. We treat S11 as the nominal MW extinction map when simulating photometry and during light-curve fitting under several alternative extinction assumptions, including CSFD, Planck, DESI *gr*, DESI *rz*, Mudur 6, and Mudur 15. Other maps created by tracing Hydrogen I emission were considered but ultimately not used in this study, the reasons for which are discussed in §3.1.

2.2. SALT Model

The SNANA framework fits observed supernova light curves using an assumed spectrophotometric model (Kenworthy et al. 2021). One such model, the Spectral Adaptive Light-curve Template 3 (SALT3), describes the rest-frame spectral energy distribution (SED) of SNe Ia as a function of phase and wavelength (Kenworthy et al. 2021). SALT3 builds upon the original SALT and SALT2 frameworks (Guy et al. 2005, 2007). SALT3 defines the flux of a SN Ia as:

$$F(p, \lambda) = x_0 [M_0(p, \lambda) + x_1 M_1(p, \lambda)] \exp[c CL(\lambda)] \quad (1)$$

where p is the rest-frame phase relative to the maximum and λ is the rest-frame wavelength. The output model parameters M_0 , M_1 , and CL are determined through the training process. M_0 represents the mean SED of a set of SN Ia and M_1 describes the time-dependent variability of the population (Kenworthy et al. 2021). CL is the wavelength-dependent color law output by the model that describes how SN Ia color impacts the SED. The model parameters are constrained by the full supernova training sample, while the SN parameters x_0 , x_1 , and c are allowed to vary independently for each observed SN. These SN parameters are then utilized in the Tripp equation (Tripp 1998):

$$\mu = m_B - M + \alpha x_1 - \beta c, \quad (2)$$

where $m_B = -2.5 \log_{10}(x_0) + C$ is the overall fitted flux amplitude (x_0) which is converted into a magnitude (m_B), the shape or stretch of the light curve (x_1), and the SN Ia color (c).

SALT training iteratively fits the SN-specific and global model parameters. This iterative process requires additional input parameters for each training sample object. Examples of input parameters are redshift, photometric calibration, the assumed MW extinction at the location of observation, and so on. The MW dust map is essential for SALT3 model training as changing $E(B - V)$ changes the foreground extinction attributed to the training observations and can therefore propagate into the recovered components. For more details on the SALT training process, we refer the reader to Taylor et al. (2023).

We perform all retraining using the open-source Python package SALTShaker (Kenworthy et al. 2021)¹.

2.3. Milky Way Dust Maps

¹ <https://github.com/djones1040/SALTShaker>

Map	Method	Scaling rel to SFD	Sky Coverage (deg ²)	Mean $E(B - V)$	$\sigma E(B - V)$
Schlafly+11	Thermal Emission	0.86	41253	0.266	1.071
CSFD	Thermal Emission	0.86	41253	0.309	1.236
Planck	Thermal Emission	0.86	41253	0.361	2.137
DESI <i>gr</i>	Stellar Reddening	0.86	13844	0.049	0.035
DESI <i>rz</i>	Stellar Reddening	0.86	12778	0.052	0.037
Mudur 6	Stellar Reddening	0.86	19922	0.054	0.058
Mudur 15	Stellar Reddening	0.86	20961	0.051	0.055
Lenz	Hydrogen I Emission	0.884	10132	0.021	0.008
Cheng	Hydrogen I Emission	0.884	16342	0.024	0.010

Table 1: Summary of the dust extinction maps used in this analysis, in their original (unfilled) form. *Method* indicates how each map estimates reddening: thermal dust emission, stellar reddening, or HI emission. *Scaling relative to SFD* is the multiplicative factor applied to all $E(B - V)$ values in the map to place it on the S11 calibration scale. *Sky Coverage* is the native (unfilled) area of the sky for which the map provides valid pixels. *Mean $E(B - V)$* and *$\sigma E(B - V)$* are computed over these native, valid pixels only. Map shorthands: CSFD to Chiang (2023); Planck to Planck Collaboration et al. (2014); DESI *gr* and *rz* to Zhou et al. (2025); Mudur 6 and Mudur 15 to the 6.1' and 15' FWHM point-spread-function maps of Mudur et al. (2022); Lenz to Lenz et al. (2017); and Cheng to Cheng et al. (2025).

We investigate a collection of MW dust maps that are produced from a variety of largely independent approaches summarized in Table 1. Our baseline MW map is S11 which is an 86% scaling of Schlegel et al. (1998) and we treat it as our truth for the simulation. While this map has been widely used in the field of SNe Ia cosmology, Chiang & Ménard (2019) found that Far-Infrared maps such as Schlegel et al. (1998) are noticeably contaminated by the Cosmic Infrared Background (CIB). Due to this contamination, extinction corrections made using the rescaled SFD, S11, map are impacted by Large Scale Structure (LSS) signals.

As potential systematic uncertainties, we include: CSFD (Chiang 2023) a corrected version of Schlegel et al. (1998) that aims to eliminate the impact of contamination from background large-scale structure, the (Planck Collaboration et al. 2014) thermal emission map, two HI emission maps: (Cheng et al. 2025) and (Lenz et al. 2017), and multiple stellar reddening maps which include the two DESI bands' maps (*rz* and *gr*) (Zhou et al. 2025), and two Mudur maps (with point spread functions 6' and 15') (Mudur et al. 2022). All of these maps are normalized relative to S11 using the scaling factors given in Table 1.

Many of these maps suffer from significant gaps in sky coverage. For this reason we develop a ‘map-filling’ process to estimate the impact of systematic uncertainties, though we note that this should not be used for precision cosmology corrections.

We complete our map pool with two validation maps. The first of these is essential for properly implementing the widely used Milky Way dust scaling systematic,

later referred to as the “historical method”. This map is simply a 5% down-scaling of our baseline S11 map. The second validation map is exclusively used to represent a non-physical extinction map in our low redshift dust map discrimination process detailed in §4.2. Each pixel in this map is random noise with an $E(B - V)$ range from 0 to 0.3, matching the clipped extinction range of our final masked maps.

3. METHODOLOGY

In this section, we describe the process of querying a representative set of modern MW dust maps, extending incomplete sky coverage maps with our own method, retraining SALT3, and propagating each map consistently through every stage of a simulated SN Ia cosmology analysis: simulating a dataset and bias corrections, light-curve fitting, systematic covariance, and cosmology fitting.

3.1. Map Reconstruction

Supernova cosmology samples are typically restricted to relatively low Galactic extinction, commonly requiring $E(B - V)_{\text{MW}} \lesssim 0.3$. Consequently, the lack of coverage near the Galactic plane in several of the maps considered here is not itself a limitation for SN cosmology to date. However, some maps still do not span the full range of low-extinction sight lines used by current and future SN samples. This becomes particularly important looking forward to LSST, which will discover SNe Ia over a much larger fraction of the southern sky. To apply these maps consistently, without restricting the analysis to their common footprint, we develop a statis-

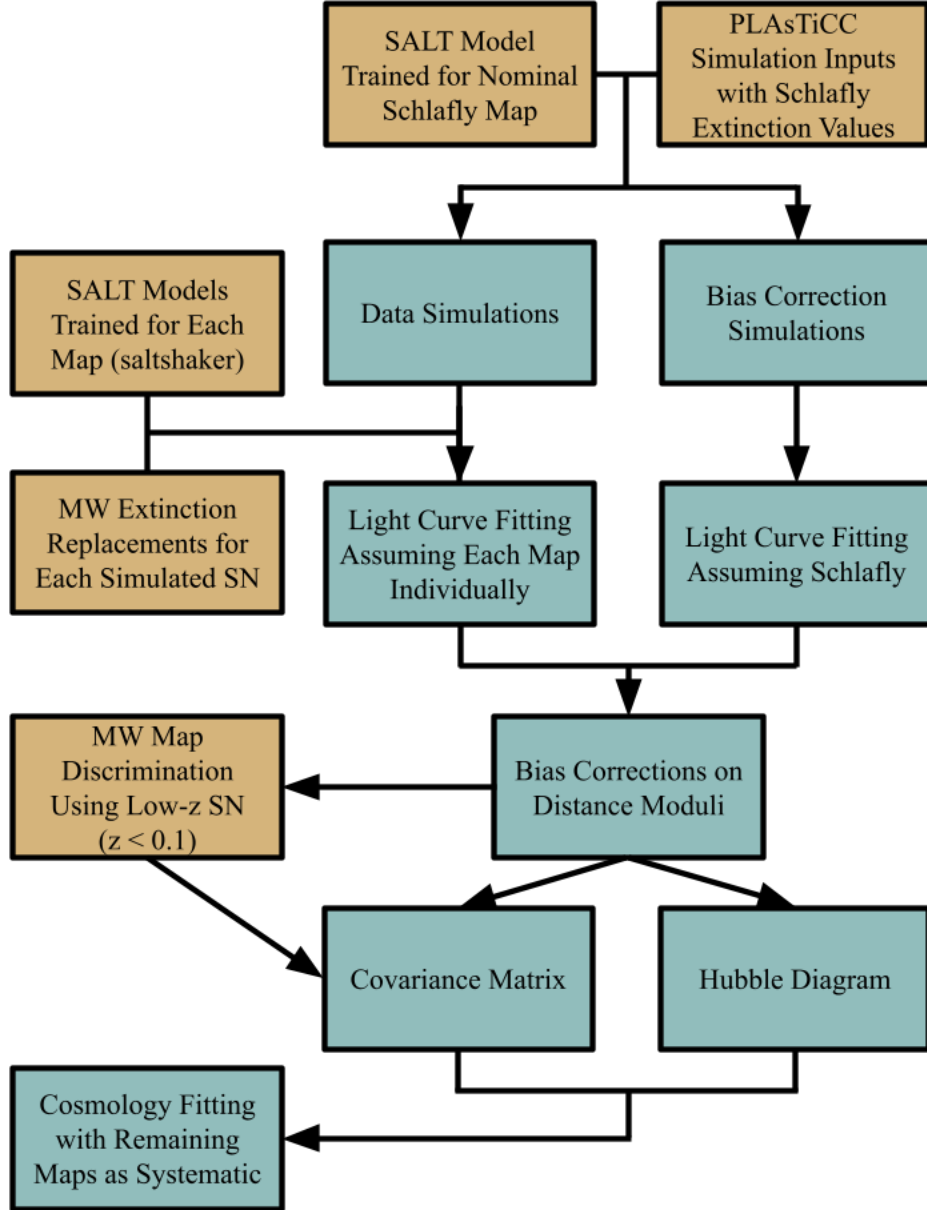


Figure 1: Workflow diagram showing how the extinction maps introduced in § 2.3 are propagated through our analysis. Stages of the SNANA framework are shown in blue, while stages specific to our analysis are shown in yellow. All maps used in this workflow are those output by the map-filling process described in § 3.1.

tical procedure for extending the incomplete maps into uncovered regions while preserving the measured properties of each map. These filled regions should not be interpreted as precision extinction corrections suitable for direct use in cosmological inference; rather, they allow us to preserve representative survey sight lines following the planned LSST WFD strategy and assess how the systematic differences of the maps propagate into SN distances and cosmological constraints.

In order to fill between the existing pixels of each map, we use S11 as a baseline and model the statistical relationship between each partial map and S11 in regions of overlapping coverage. In regions where we have both S11 and one of the coverage-lacking maps, we build an extrapolation scheme to fill in the missing areas in the latter. We do not use nearest-neighbor filling or bilinear interpolation as doing so would rely on small-scale local pixel groups informing the filling of pixels large angular distances away. It would also largely fail to reproduce

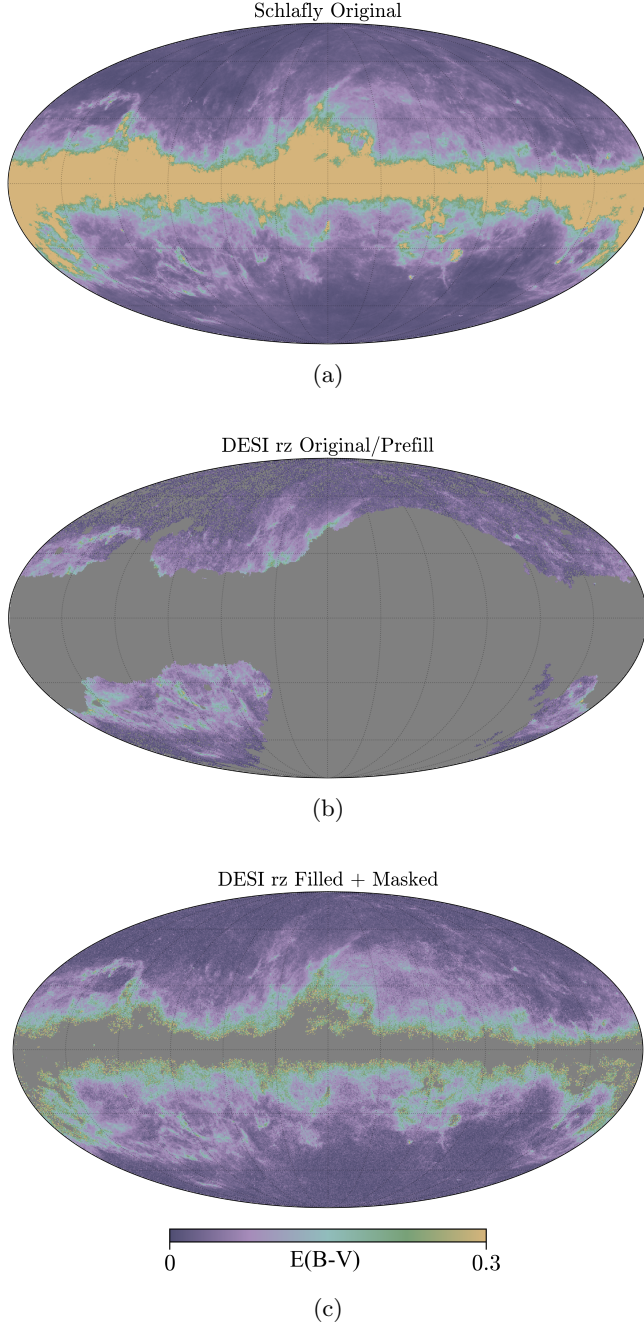


Figure 2: The DESI rz map-filling process, shown in Galactic coordinates with a shared color scale. Panel (a) shows the $S11$ reference map used to fill gaps in the original DESI rz map shown in panel (b). Panel (c) shows the resulting filled map, with high-extinction regions masked to exclude poorly constrained areas with limited native coverage. This final map is used throughout the analysis, and the same process is applied to all dust maps. See § 3.1 for details.

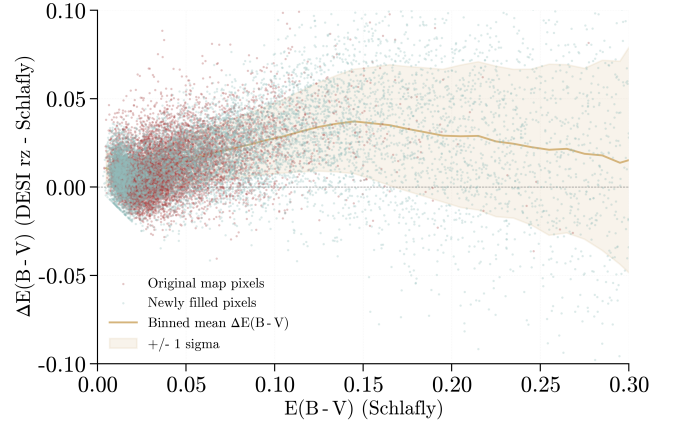


Figure 3: The $E(B-V)$ residual between DESI rz and the co-located $S11$ pixel as a function of its $E(B-V)$ value, shown for a random subsample of original (red) and filled (blue) DESI rz pixels. The yellow line shows the binned mean residual of the original pixels, with the shaded region indicating the corresponding 1σ Gaussian width in each bin. Together, these define the offset relation used to generate the filled pixels shown in Figure 2 panel (c). DESI rz is shown as a representative example of the equivalent validation performed for each filled map.

galactic band structure as none of the partial maps we investigate initially extend into these regions. Gaussian processes were not well suited for this purpose either as local pixels are not the best prior for filling logic and they fail to recreate realistic galactic structure. Rather, the $S11$ value at any given location acts as a much better informer, preserving the existing map versus $S11$ relation and providing a sky location-based prior. Our objective is not to reconstruct the true dust field, but to preserve the observed statistical relationship between each map and $S11$.

We reconstruct missing pixels by first calculating the binned average residual between a given map and $S11$ as a function of $E(B-V)$ in $S11$, equation shown below. Residuals are computed only for pixels where both maps have valid measurements. Invalid pixels are masked before this process to ensure the relation is only being trained on real data.

$$\Delta E(B-V) = E(B-V)_{\text{map}} - E(B-V)_{\text{Schlafly}+11} \quad (3)$$

Binning by $S11$ extinction allows the residual distribution to vary with extinction, capturing the systematic dependence of map differences on dust column density. We choose to use equal width binning as population size is not a typical barrier for large maps and equal population approaches tends to cause extremely wide $S11$

$E(B - V)$ space binning at the highest and lowest extinction values. A Gaussian distribution is then built, informed and centered on this residual versus $E(B - V)$ trend. One Gaussian is fit per bin, for which the mean and standard deviation are measured directly from the existing pixel residuals. We adopt a Gaussian approximation to preserve the mean and standard deviation of the residual distribution rather than reproduce every individual fluctuation.

For each missing pixel in a given map, we query the $E(B - V)$ value of S11 and the bin in which that pixel falls in S11 $E(B - V)$ space to randomly draw a residual from the Gaussian. This residual is then added to that pixel locations' S11 extinction value and this $E(B - V)$ is returned as the extrapolated filled pixel.

$$E(B - V)_{\text{fill}} = E(B - V)_{\text{Schlafly} + 11} + \sim N(\Delta E(B - V)_z, \sigma_z) \quad (4)$$

To avoid regions that are poorly constrained, we mask out high-extinction sight-lines with $E(B - V) > 0.3$ (e.g., Smith et al. 2020).

A single filled realization is generated for each dust map. These filled maps support realistic LSST supernova simulations using empirical dust data.

We validate our filled maps by inspecting the extinction trends of the filled and original pixels for each map as shown in Figure 3. By comparing a sample of each population of pixels, we confirm that the extrapolated pixel values follow a distribution similar to that of the original map pixels in the baseline map $E(B - V)$ space. For the DESI rz map, the final filled and masked map retains the overall characteristics of the original, with the mean $E(B - V)$ increasing by 0.0244 mag and the standard deviation in the dust values increasing by 0.0305 mag. Such relations can be seen for each map used in this analysis in Tables 1 and 2. Panel c of Figure 2 shows the large-scale galactic structure that is preserved in our final filled and masked extinction maps.

Our map-filling pipeline is based on a few key assumptions. The foundation for this process is the premise that the map - S11 extinction relationship measured in the overlapping regions is representative of the missing regions. We also assume that the residual relation depends primarily on the extinction value given by S11. Finally, we work under the premise that a Gaussian distribution is capable of adequately describing this residual relation, preserving the mean and variance of the trend. We only fill pixels of low S11 extinction in order to reduce sensitivity to the least constrained extrapolations.

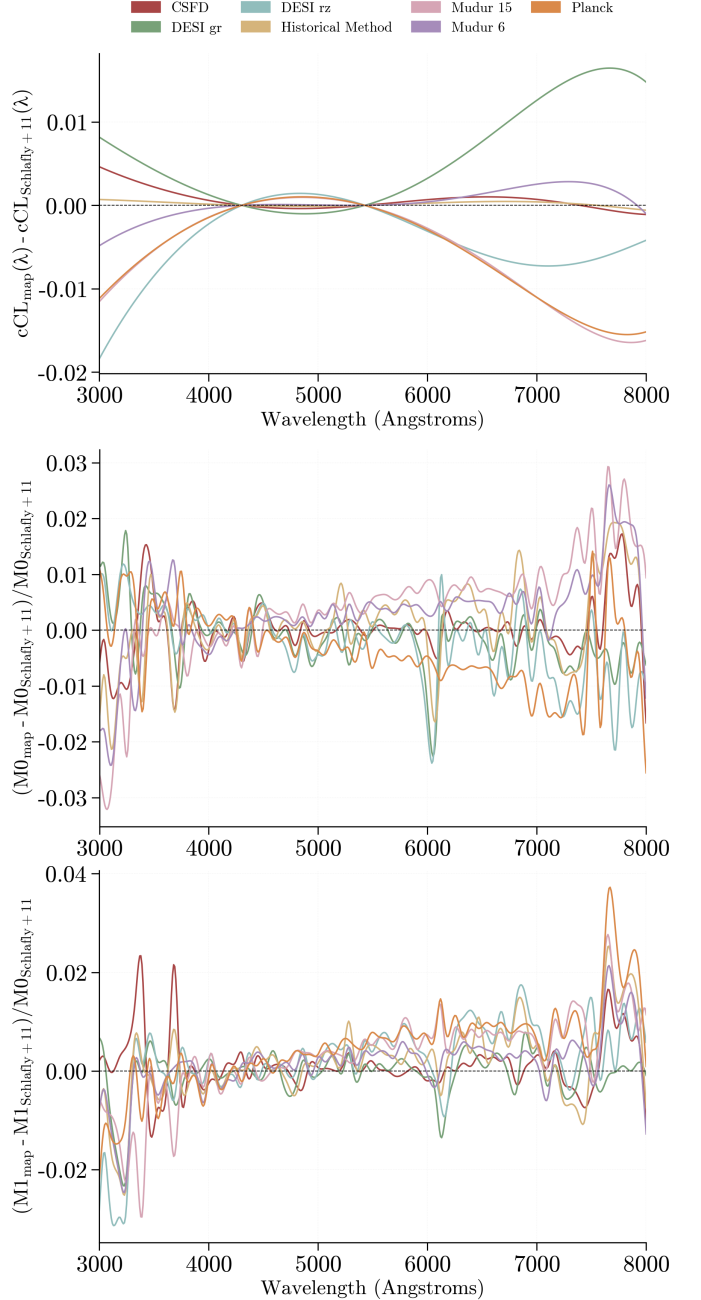


Figure 4: The SALT3 color law and M_0/M_1 surfaces, evaluated at peak phase, for each retrained model used in this work. All panels show residuals relative to the nominal S11 baseline model and share the legend above panel a). Panel a) shows the color law (CL), panel b) the M_0 surface representing the mean SN Ia SED, and panel c) the M_1 surface describing population variability, as described in § 2.2. All models use the same training set, differing only in the Milky Way extinction values assigned to each training SN.

We note that the filling process is poorly constrained for the Hydrogen emission maps, (Lenz et al. 2017) and (Cheng et al. 2025). These maps have native $E(B - V)$ ranges with maximum values below 0.05, making extrapolation well beyond this range unreliable. We explored implementing these maps in the SALT3 retraining stage exclusively but ultimately decided against it as doing so put heavy constraints on our training set. In the lowest extinction parts of the sky, the other maps used in this analysis are representative of the difference between the HI maps and our nominal. However, the HI maps report consistently lower extinction values than S11. The regions of moderate extinction per S11 are where the HI maps provide the most unique perspective on Galactic extinction, further motivating future work to expand the SALT3 training set into lower extinction regions of the sky as is discussed in §4.1. Ultimately, the HI maps provide an interesting independent measurement of galactic dust, but do not offer sufficient sky coverage to be used in this analysis.

3.2. SALT3 Retraining

To ensure each MW dust map is propagated consistently throughout the supernova analysis pipeline, we retrain the SALT3 spectral model using the same extinction assumptions we later adopt during light-curve fitting. Replacing only the Milky Way reddening values assigned during light-curve fitting would leave the SALT3 model itself trained under a different extinction map assumption, producing an internally inconsistent analysis. We retrain a separate SALT3 model for every dust map considered in this work, to ensure that the training and fitting stages always adopt the same MW extinction map.

All models are retrained using SALTShaker with an identical configuration, optimization procedure, quality cuts, and underlying training sample. The initial training sample is that of the published SALT3.DOVEKIE model Popovic et al. (2026). This sample includes data from Hamuy et al. (1996), Riess et al. (1999), Jha et al. (2006), Hicken et al. (2009), Conley et al. (2011a), Hicken et al. (2012), Krisciunas et al. (2017), Foley et al. (2018), Sako et al. (2018), Scolnic et al. (2018), Brout et al. (2019). Holding the training procedure fixed ensures that the adopted MW extinction map is the only intentionally varied input between retrained models.

We use the filled dust maps described in § 3.1 for retraining to preserve as much of the SALT3 training sample as possible. Regions with $E(B - V)_{\text{Schlafly+11}} > 0.3$ in the filled maps are excluded. For each map, we replace the MW $E(B - V)$ values for all SNe Ia with valid data. Some training SNe Ia lie in regions for which a

valid $E(B - V)$ value cannot be obtained for one or more of the dust maps. Any SNe Ia lacking a valid extinction value from even a single map are removed from the training sample for every map. These requirements remove 8 objects, leaving a common sample of 1075 SNe Ia used for all model retrains.

All SALT3 model retrains converge successfully under a convergence tolerance of 10^{-4} . A summary of training statistics for all maps can be seen in Table 3. This summary includes for each map; the number of training SN cut from our final sample due to insufficient coverage from that map, the final training Chi2 statistic, and the extinction value range of the training SN. After built-in SALTShaker training cuts (Kenworthy et al. 2021), all of the models train on the same 1050 SNe Ia. In Figure 5, we visualize the SALT3 training set and per-map extinction distribution.

Each retrained SALT3 model is subsequently paired with the corresponding MW extinction map during the light-curve fitting stage, as described in section 3.3, such that every fitting option represents one self-consistent implementation of a single dust map.

3.3. Light-Curve Fitting and Map Implementation

After retraining SALT3 models for each dust map, we performed light-curve fitting using the corresponding extinction assumptions. Following the simulation stage, we feed the light-curve fitting stage with new assumed extinction values for each simulated SN for each map. We treat each map identically, simultaneously replacing the SALT model used and the assumed MW extinction correction used for each of our simulated supernovae. Using these extinction assumptions overrides just before the light-curve fitting stage allows the same simulated data set to be fit under different MW extinction assumptions without regenerating the simulations.

Each fitting option represents a complete extinction map implementation, including both the SALT model and individual supernova dust value replacements. The same iteration of the filled dust map used for SALT retraining is used for all aspects of the fitting process. All other fitting parameters are held fixed across the map-dependent light-curve fits.

In addition to the two-part map fitting implementations, we perform fits for which the SALT3 retraining and individual SN extinction corrections are varied independently. For such fitting options, we use the nominal S11 fitting parameters, replacing either the SALT model or the per-SN extinction corrections of a given map. These configurations allow us to distinguish changes caused by these two map implementations individually

Map	Filled Coverage	Filled mean	Filled STD	Masked	Final Coverage	Final mean	Final STD
Schlaflly+11	0	-	-	7350.2	33902.7	0.0689	0.0642
5percent	0	-	-	7080.3	34172.7	0.0673	0.0640
CSFD	0	-	-	7351.1	33901.9	0.0689	0.0642
Planck	0	-	-	6988.4	34264.5	0.0786	0.0672
DESI <i>gr</i>	23348.5	0.1350	0.1290	2665.1	34527.7	0.0756	0.0673
DESI <i>rz</i>	24414.1	0.1309	0.1316	2537.8	34655.0	0.0764	0.0675
Mudur 6	17311.4	0.1074	0.0966	1124.7	36109.0	0.0663	0.0652
Mudur 15	16272.3	0.1110	0.0959	1043.8	36189.9	0.0652	0.0658

Table 2: Diagnostics of the gap-filling and masking procedure applied to each map. *Filled Coverage* is the sky area newly assigned $E(B - V)$ values by the filling procedure (i.e. pixels that were missing in the original map). *Filled mean* and *Filled STD* are the mean and standard deviation of $E(B - V)$ over only these newly filled pixels. *Masked* is the sky area removed by the final masking threshold of $E(B - V) > 0.3$ mag. *Final Coverage*, *Final mean*, and *Final STD* give the corresponding statistics for the map after both filling and masking have been applied. All coverage and masked columns are in deg^2 ; mean and STD columns are in $E(B - V)$ (mag). Schlaflly+11, 5percent, CSFD, and Planck required no synthetic filling and so have no filled-column entries; 5percent is a constructed systematic map defined as a $0.95 \times$ scaling of S11, representative of the historical 5% calibration systematic used in the field.

Map/Model	SALT3 χ^2	$E(B - V)$ min	$E(B - V)$ max	$E(B - V)$ mean	$E(B - V)$ std	Num SN cut
Schlaflly+11	74465.95	0.0045	0.2697	0.0324	0.0297	7
5%	74227.83	0.0043	0.2562	0.0308	0.0282	7
CSFD	74568.14	0.0048	0.2701	0.0322	0.0296	8
Planck	74332.31	0.0069	0.2549	0.0384	0.0311	4
Desi <i>gr</i>	74136.47	0.0	0.2550	0.0348	0.0309	6
DESI <i>rz</i>	74171.80	0.0	0.2937	0.0365	0.0327	6
Mudur 6	74424.83	0.0	0.1914	0.0305	0.0251	4
Mudur 15	74446.26	0.0	0.2422	0.0276	0.0238	3
Noise Map	76079.34	0.0	0.2998	0.1458	0.0253	0

Table 3: Diagnostics for the SALT3 light-curve models retrained using each dust map. *SALT3 χ^2* is the total training χ^2 reported by the SALT3 fitter; it serves as a diagnostic of successful convergence for each retraining run rather than a measure of relative fit quality across maps. $E(B - V)$ min, max, mean, and std describe the distribution of Milky Way extinction values substituted for the training supernovae prior to retraining, with each supernova’s $E(B - V)$ drawn from the corresponding map at its sky position. *Num SN cut* is the number of training supernovae excluded from the retraining sample because the map lacked $E(B - V)$ coverage at that supernova’s location.

and compare how each contributes to the SN distances recovered in the full two-part map fitting.

To isolate the impact of the adopted Milky Way dust map, subsequent comparisons are restricted to SN successfully fit in every fitting configuration. This prevents differences in sample composition from influencing comparisons between dust maps. This common SNe selection philosophy is applied at all stages throughout our analysis.

3.4. Cosmology Propagation

The parameters produced by the light-curve fitting stage as described above are propagated through the

standard SN Ia cosmology analysis pipeline to obtain bias-corrected distance moduli and constraints on cosmological values w_0 , w_a , and Ω_m .

Before cosmology fitting, we correct the recovered SN Ia distance moduli for biases using the BEAMS with Bias Corrections (BBC) method (Kessler & Scolnic 2017). This process requires a large dataset of SNe, simulated and processed assuming the baseline S11 extinction map, that is otherwise simulated and fit with identical parameters to those applied to the data simulation set.

We investigate whether dust maps must be propagated into these bias-correction simulations, or a single

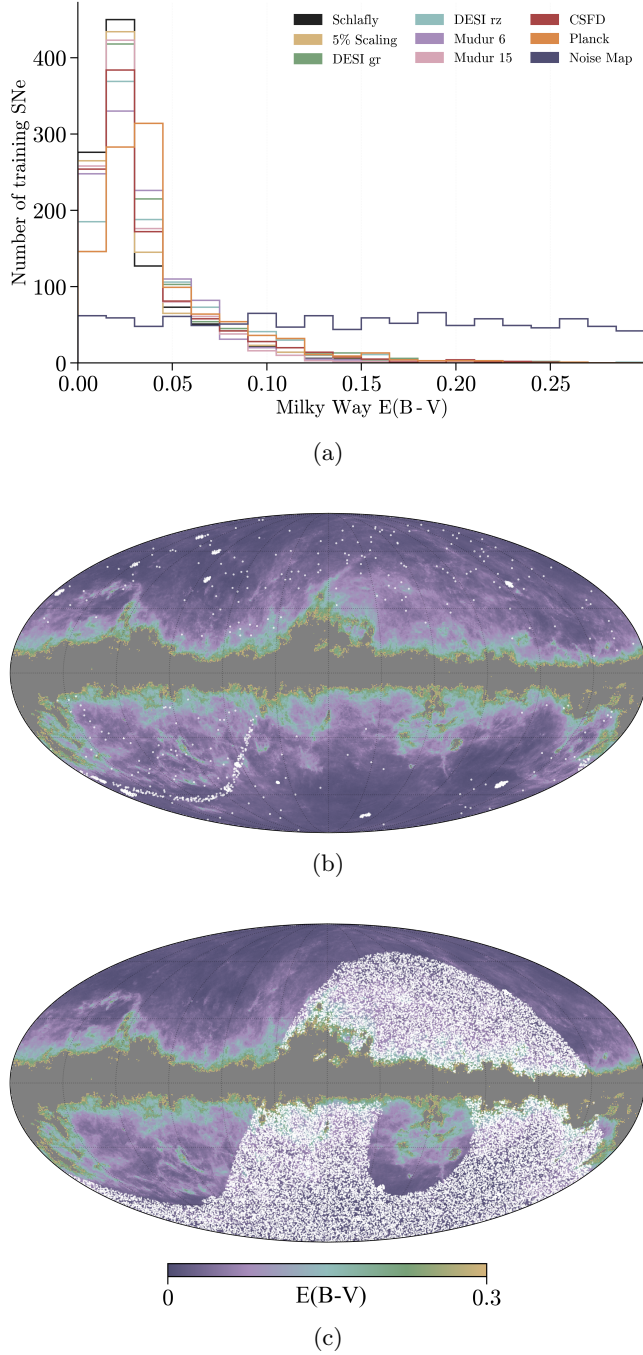


Figure 5: SALT3 model retraining diagnostics. Panel (a) shows the distribution of extinction values used to retrain the SALT3 models shown in Figure 4. Panel (b) shows the final SALT3 training sample of 1075 SNe overlaid on the baseline S11 Milky Way extinction map. Panel (c) shows the final LSST low- and high-redshift simulation samples used in this analysis overlaid on the same map. Panels (b) and (c) share the color scale shown in panel (c).

S11 extinction map can be used for all scenarios. We test the effect of this process on the DESI rz map which produces the largest changes in distances modulus relative to S11 when the MW dust systematic is introduced only through SALT3 retraining and light-curve fitting. We find that the implementation of this systematic in the bias correction samples has a negligible impact on the resulting supernova distance residuals as seen in figure ??, changing the overall Hubble scatter by less than 0.001 mag. We therefore do not generate independent bias-correction simulations for every dust map in our primary analysis and leave a more comprehensive investigation of dust-map-dependent bias corrections to future work.

For each dust map configuration (in training and fitting), the resulting distance modulus shifts are propagated into a systematic covariance matrix used for cosmological fitting and we follow the standard approach in the field, described in detail by (Conley et al. 2011b), and utilize re-binned covariances and Hubble diagram (Brout et al. 2021; Kessler et al. 2023) due to our very large LSST sample sizes.

Finally, we fit the resulting Hubble diagram and covariance matrices using **WFIT**², an alternative to the traditional Markov Chain Monte Carlo’s approach that uses a simple grid-based χ^2 approach to fit for cosmological parameters. In this work we fit a flat $w_0 w_a$ CDM model and employ a Cosmic Microwave Background (CMB) prior which is further discussed in § 4.3.

4. RESULTS AND DISCUSSION

Following Peterson et al. (2022), we use the low-redshift SN sample to test whether alternative MW dust maps produce distinguishable changes in the Hubble diagram. Restricting this test to $z < 0.1$ minimizes the dependence of the comparison on the assumed cosmological model and allows the maps to be evaluated primarily through their effect on SN distance estimates. Maps producing sufficiently large increases in Hubble-diagram scatter relative to the nominal S11 map are excluded from the subsequent cosmological analysis.

Following this, the remaining dust maps are propagated through the full low- and high-redshift sample and treated collectively as a Milky Way extinction systematic. We compare the resulting cosmological constraints with those obtained using the conventional 5% scaling systematic of the S11 map. This allows us to determine both whether current SN samples can distinguish among modern Galactic extinction maps and whether

² <https://github.com/RickKessler/SNANA/blob/master/src/wfit.c>

map choice represents a significant source of uncertainty for SN Ia cosmology.

4.1. Impact of SALT3 Retraining

This work follows a two method approach by propagating the MW extinction systematic through both supernova light-curve fitting and SALT model training. In past SN Ia analyses, the SALT model training aspect of Galactic dust has been overlooked. An important point of investigation for this study has been determining if per-SN extinction correction replacements are sufficient for a more comprehensive dust map systematic.

To answer this question, we carry each Galactic dust map implementation through two additional light-curve fitting options, each independently replacing the extinction corrections using only one of the two methods as described in §3.3. Throughout this analysis, we consistently find a given dust maps SALT model implementation to have the dominant effect on the maps resulting distance modulus residuals. This distinction can be seen in Fig 6 for the DESI rz map. This trend in binned distance modulus residuals is consistent for each of our maps; salt “training” causes larger residuals than per-supernovae “fitting”. For the case of the DESI rz map, including SALT model retraining increases the scatter in the high-redshift distance-modulus residuals by approximately 0.005 mag, with the RMS increasing from 0.0092 to 0.0139 mag above $z_{HD} = 0.6$ and from 0.0104 to 0.0158 mag above $z_{HD} = 0.8$. Additionally, shown by Fig 6, is the impact of filter loss for high redshift supernova in the data set. The vertical lines represent the redshifts at which the g and r filters drop out of SN photometry, coinciding with increased fluctuations in the distance modulus residuals and highlighting the importance of multi-band photometry for high-redshift SN Ia studies.

The comparatively small impact of replacing the extinction correction only during light-curve fitting can be understood through its effect on the fitted SALT color c . Figure 7 shows the change in c as a function of the difference in $E(B - V)$ between DESI rz and the nominal S11. We find a strong anti-correlation between these quantities, with the binned relation closely following $\Delta c = -\Delta E(B - V)$. This suggests that for a given SN, a decrease in the assumed Galactic extinction produces an increase in the color attributed to that SN by the light-curve fitting process. Consequently, the per-SN MW extinction correction replacements done exclusively in the light-curve fitting stage are absorbed by the resulting color parameter. This can be seen in the lower panel of Figure 7, which shows that the resulting distance-modulus shifts after light-curve correc-

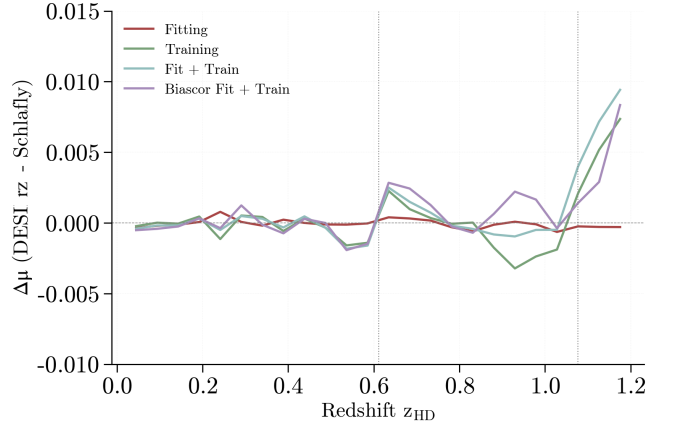


Figure 6: Binned Hubble diagram for four implementations of the DESI rz extinction map. The red “Fitting” curve replaces the extinction corrections for individual SNe while retaining the baseline S11 SALT3 model. The green “Training” curve uses the DESI rz retrained SALT3 model while retaining the S11 extinction corrections. The blue “Fit + Train” curve applies both replacements. The purple curve additionally uses a bias-correction simulation generated with the DESI rz extinction map, as described in § 3.4. Its agreement with the “Fit + Train” curve motivates the use of the S11 bias-correction sample for the primary analysis. The vertical lines mark the redshift at which the g band ($z = 0.6248$) and r band ($z = 1.0672$) drop out of the photometry.

tions are substantially smaller than the naive assumption of $\Delta\mu = -3.1\Delta E(B - V)$. This is consistent with the fitted color absorbing much of the change in the assumed Galactic reddening. A residual trend in distance modulus does remain due to differences between the Galactic color law and SALT SN color law, and this is the level systematic effect that modern surveys have reported (e.g. Vincenzi et al. 2024) in current cosmological error budgets.

These results demonstrate that modifying only the per-SN extinction corrections does not capture the full impact of Galactic dust-map uncertainty. We therefore find that propagating each dust map through SALT3 model training is necessary in addition to its implementation during light-curve fitting. Unless otherwise specified, all subsequent results use the self-consistent fit+train implementation, in which the same Galactic dust map is adopted during both SALT3 training and light-curve fitting.

4.2. Discriminating Dust Maps with the Low- z Sample

We use our simulated low-redshift SN sample to identify dust maps that produce poorer fits to the Hubble

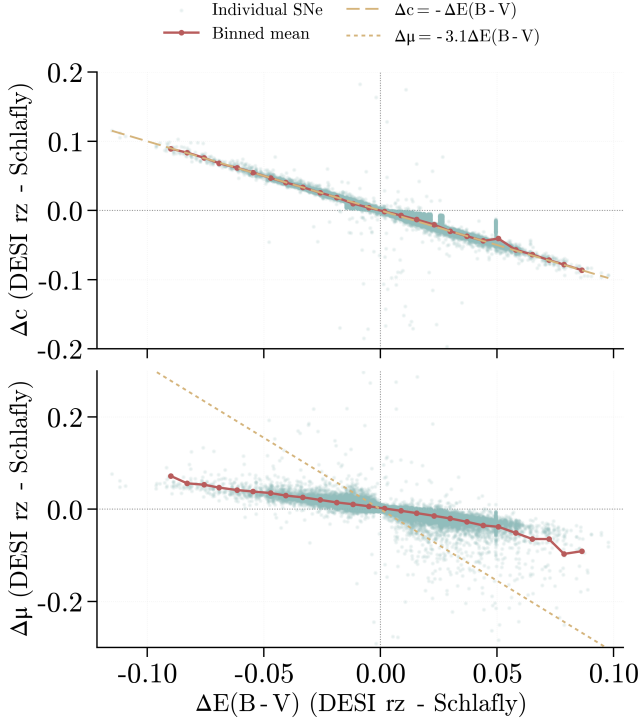


Figure 7: Residual relationships for the DESI rz light-curve “Fitting” only extinction implementation relative to S11. Panel (a) shows the $E(B-V)$ residual as a function of color residual, with the binned mean relation (red) and $\Delta c = -\Delta E(B-V)$ relation (yellow). Panel (b) shows the $E(B-V)$ residual as a function of distance modulus residual, with the binned mean relation (red) and $\Delta \mu = -3.1\Delta E(B-V)$ relation (yellow).

diagram than our baseline S11 map. Restricting this test to the low- z sample allows us to evaluate differences in Hubble diagram scatter while minimizing sensitivity to the assumed cosmological model. For each dust map, we use the bias-corrected distance moduli to calculate the Hubble residuals and their corresponding X^2 as shown below.

$$X^2 = \sum_i \left[\frac{\mu_i - \mu_{\text{mod}}(z_i) - \langle \mu - \mu_{\text{mod}} \rangle_w}{\sigma_{\mu,i}} \right]^2. \quad (5)$$

Here, μ_i and $\sigma_{\mu,i}$ are the bias-corrected distance modulus and its uncertainty respectively, and $\mu_{\text{mod}}(z_i)$ is the distance modulus predicted by the reference cosmological model. The weighted mean Hubble residual, $\langle \mu - \mu_{\text{mod}} \rangle_w$, is subtracted so that the statistic is sensitive to differences in Hubble-diagram scatter rather than an overall offset in distance modulus. Following Peterson et al. (2022), we refer to this statistic as X^2 rather than χ^2 and we make no assumption about its distribution under a null hypothesis.

We construct μ_{mod} using the best-fit cosmology recovered from the nominal S11 sample. We assume a flat Λ CDM cosmology with $w = -1$ and fit for Ω_m , after which the best-fit S11 value of Ω_m is held fixed when evaluating all alternative dust maps. This ensures that every map is compared against the same reference Hubble diagram. We repeat this calculation using the independently fitted Ω_m for each dust map and find no meaningful change in the relative ordering of the maps, consistent with Peterson et al. (2022).

Using the same μ_{mod} , we also calculate the RMS of the Hubble diagram scatter. The RMS provides a complementary measure of map performance that does not depend on the uncertainties assigned to the individual SN distances. We define the RMS as:

$$\text{RMS} = \sqrt{\frac{1}{N} \sum_i [\mu_i - \mu_{\text{mod}}(z_i) - \langle \mu - \mu_{\text{mod}} \rangle]^2}. \quad (6)$$

When calculating X^2 using the distance modulus uncertainties obtained independently for each dust map, we find that several alternative maps produce values below that of the nominal S11 analysis. This behavior is unexpected because S11 is the input extinction map of the simulation and should therefore provide the most consistent recovery of the simulated SN distances. We find that the improved X^2 values arise from a map-dependent change in the reported distance uncertainties rather than from a reduction in the Hubble-diagram scatter. As discussed in § 4.1, changes in the assumed Milky Way extinction propagate into the fitted SALT color. Maps that preferentially produce redder fitted SN colors yield larger distance-modulus uncertainties, increasing the denominator of X^2 and artificially reducing its value even when the Hubble-diagram scatter increases.

To isolate differences in the recovered distances from this map-dependent change in uncertainty, we additionally calculate X^2 using the nominal S11 uncertainty for each SN Ia while retaining the distance modulus recovered with each alternative map. Figure 8 compares the resulting X^2 residuals with those obtained using the map-dependent uncertainties. For most maps that initially yield X^2 below that of S11, adopting the corresponding S11 uncertainties shifts the residual to positive values. The agreement between this fixed-error comparison and the RMS provides a more direct measure of the additional Hubble-diagram scatter introduced by each alternative extinction map.

We therefore use the low-redshift Hubble diagram scatter to determine which Galactic dust maps remain viable for inclusion in the Milky Way extinction sys-

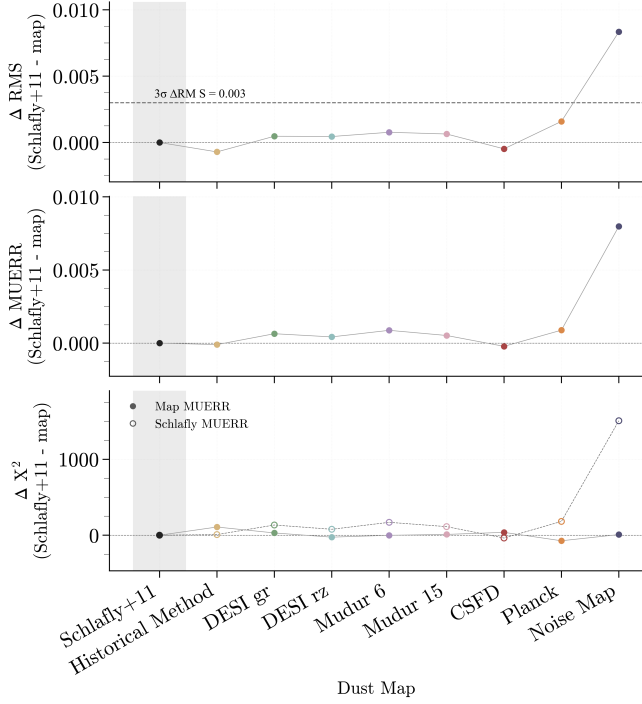


Figure 8: Comparison of Hubble-diagram scatter and distance-modulus uncertainties for each Milky Way dust map relative to the nominal S11 implementation. Panel (a) shows the residual RMS Hubble-diagram scatter with the 3σ threshold line shown at $\Delta RMS = 0.003$. Panel (b) shows the residual mean distance-modulus uncertainty (MUERR) for each map which is used in panel (c) to plot the residual X^2 with either the corresponding MUERR (closed) or the S11 MUERR (open).

tematic. Figure 8 shows the residual X^2 and RMS of each map relative to S11. Our final Hubble diagram has a RMS of approximately 0.13, which when propagated through the following relation with our sample size of 11,212 SNe Ia, gives us an intrinsic scatter floor 1σ deviation of approximately 0.001.

$$\sigma(RMS) = \frac{RMS}{\sqrt{2N}} \quad (7)$$

Using this approach, any maps which produce a residual $RMS > 0.003$ (which corresponds to $3\sigma \Delta RMS$) when compared to S11 could be excluded from the subsequent cosmological analysis, while the remaining maps are propagated as the MW extinction systematic. For the sample we use in this work, none of the maps we consider exceed this RMS threshold. However, if future datasets or improved SN standardization techniques result in lower sample RMS, this threshold could be lowered and our map discrimination process could prove more effective. This could also be achieved with a more

ambitious redshift range, extending to z of 0.17 to include our entire WFD sample.

We do find that, our random noise map differs from S11 by $\sim 9\sigma$, demonstrating that the 10 yr low- z LSST sample has the statistical power to discriminate between maps. This suggests that our map pool aligns closely with S11 thus, all six alternative maps are propagated into our final MW extinction systematic. These surviving maps are evaluated using the full WFD, and DDF SN sample extending to $z = 1.29$, as described in § 4.3.

4.3. Cosmological Impact and Systematic Uncertainty Budget

Below we present cosmology results for three sample sizes of LSST simulations. The two intermediate samples, tables 4a and 4b, consist of the Y1 and Y5 sub-sampled sets of the entire Y10 data set, results reported in table 4c. For all results across these tables we assume a CMB prior as done by Sánchez et al. (2022), for which we use a σ_{RCMB} of 0.0057. We compare the statistical constraints for each stage of the survey to two treatments of the Milky Way extinction systematic. The primary treatment utilizes our own map-based systematic, which propagates all studied extinction maps through both SALT3 training and light curve fitting. The second is the historical treatment based on a 5% scaling of the nominal S11 map following the treatment of Pantheon+ (Brout et al. 2022), the Dark Energy Survey (Smith et al. 2020), and the LSST DESC Science Requirements Document (LSST-SRD The LSST Dark Energy Science Collaboration et al. 2018b), for comparison to our new technique.

The impact of the expanded Milky Way dust systematic is already apparent in the Year 1 sample shown in Table 4a and is consistent across years. The dark energy Figure of Merit decreases from 238.0 for the statistical only analysis to 195.9 when the full dust map systematic is included, corresponding to a degradation of approximately 18%. In contrast, the historical 5% scaling produces essentially no change in the Year 1 Figure of Merit. Thus, even at early LSST statistical precision, the diversity among plausible Galactic extinction maps produces a substantially larger cosmological uncertainty than is represented in the LSST-SRD. This difference persists as the statistical power of the SN sample increases.

For the Year 5 sample in Table 4b, the Figure of Merit decreases 17% over statistical, whereas the historical treatment decreases the Figure of Merit by 5%. Similarly, for the full Year 10 sample in Table 4c, our map-based systematic results in a 15.2% reduction compared to 4.4% for the historical treatment. The cosmological impact of our more complete Milky Way extinction sys-

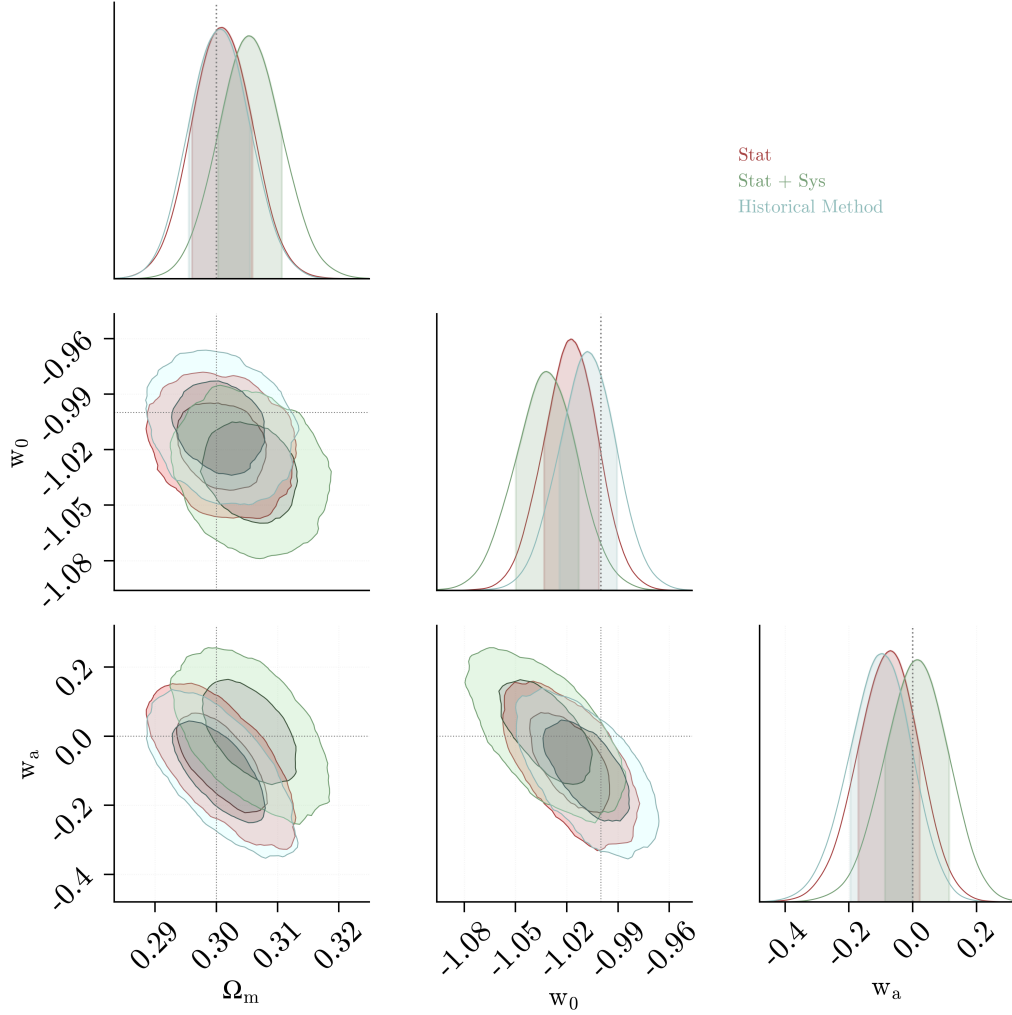


Figure 9: Cosmological results for the full 10-year LSST simulation sample. “Stat” denotes the nominal S11 cosmology, while “Stat + Sys” includes the dust maps implemented through both SALT3 training and light-curve fitting. “Historical Method” uses the 5% scaling of the nominal S11 extinction map, implemented through light-curve fitting only.

tematic is therefore more than three times larger than the previous prescription. Importantly, this fractional degradation remains approximately constant as the statistical sample grows, demonstrating that increased SN statistics alone do not substantially reduce the relative importance of Galactic extinction uncertainty.

Following Brout et al. (2022), we report our final Milky Way dust systematic in table 5 using the following relations:

$$\Delta\theta_{\text{sys}} = \theta_{\text{stat+sys}} - \theta_{\text{stat}} \quad (8)$$

$$\sigma_{\theta,\text{sys}} = \sqrt{\sigma_{\theta,\text{stat+sys}}^2 - \sigma_{\theta,\text{stat}}^2} \quad (9)$$

$$\frac{\sigma_{\theta,\text{sys}}}{\sigma_{\theta,\text{stat}}}.$$

Here, θ represents the cosmological parameter of interest and is taken to be w_0 , w_a , or Ω_m . Stat represents values obtained from the cosmological fitting of the statistical/nominal covariance matrix and hubble diagram. Sys similarly represents the results of fitting the covariance matrix that includes each of our six MW extinction maps—CSFD, Planck, DESI gr and rz, and Mudur 6 and 15—in addition to the nominal S11 map. As a comparison, we also report these values using the cosmology fitting of the 5% scaling systematic widely used in SNe Ia analyses.

Table 5 shows the Year 10 result directly in terms of the systematic contribution to each cosmological parameter uncertainty. For w_0 , the map based treatment contributes $\sigma_{w_0,\text{sys}} = 0.010$, corresponding to 71% of the statistical uncertainty, compared with 0.008, or 57%, for the historical treatment. The difference is considerably

Fit	ω_0	$\sigma\omega_0$	ω_a	$\sigma\omega_a$	Ω_m	$\sigma\Omega_m$	FoM
Stat	-0.981	0.048	-0.272	0.0219	0.303	0.006	238.0
Stat + Sys	-1.009	0.051	-0.176	0.234	0.302	0.006	195.9
Historical Method	-0.980	0.048	-0.257	0.219	0.304	0.006	238.2

(a) 1-year sample

Fit	ω_0	$\sigma\omega_0$	ω_a	$\sigma\omega_a$	Ω_m	$\sigma\Omega_m$	FoM
Stat	-0.998	0.022	-0.179	0.118	0.301	0.005	561.2
Stat + Sys	-1.015	0.025	-0.116	0.125	0.302	0.006	467.9
Historical Method	-0.987	0.023	-0.211	0.119	0.301	0.005	536.1

(b) 5-year sample

Fit	ω_0	$\sigma\omega_0$	ω_a	$\sigma\omega_a$	Ω_m	$\sigma\Omega_m$	FoM
Stat	-1.018	0.015	-0.077	0.094	0.301	0.005	853.3
Stat + Sys	-1.032	0.018	0.011	0.097	0.306	0.005	723.6
Historical Method	-1.008	0.017	-0.103	0.094	0.301	0.005	816.1

(c) 10-year sample

Table 4: Cosmological fit results from the SNANA `wfit` fitter, for the (a) 1-year, (b) 5-year, and (c) 10-year LSST samples. *Stat* gives statistical-only outputs computed using the nominal S11 calibration, with no systematics included. *Stat + Sys* additionally includes the systematic covariance matrix built from the light-curve fitting and SALT3 training outputs of each dust map. *Historical Method* instead applies the historical 5% $E(B - V)$ scaling systematic during light-curve fitting only. FoM is the Figure of Merit output by the `wfit` cosmological fitting process.

larger for w_a : the map based systematic contributes $\sigma_{w_a, \text{sys}} = 0.025$, or 27% of the statistical uncertainty, whereas the historical treatment contributes only 0.006, or 6%. Taken together, Tables 4a–5 demonstrate that the commonly adopted 5% rescaling of the S11 extinction map does not span the cosmological uncertainty associated with the diversity of modern Milky Way dust maps.

The importance of propagating the MW extinction systematic through the SALT3 training stage of type Ia supernova analysis as discussed in 4.1 is a strong motivation for future work by surveys including LSST, DESI, 4MOST and Roman. Existing target of opportunity programs could be utilized to expand salt training sets in regions of very low Galactic extinction. Creating a more robust salt training set in the low extinction sky would allow for this comprehensive dust systematic to be applied to upcoming cosmology analyses using a larger set of Milky Way extinction maps. Maps such as the Hydrogen emission maps left unused in this study could be included and the map filling process detailed in §3.1 could be removed from future dust systematic analyses

In this work, we investigate the impact of Milky Way extinction uncertainties on Type Ia supernova cosmology and develop a framework for propagating these uncertainties consistently through both SALT3 model training and light curve fitting. Using simulated LSST supernova samples, we compare a representative set of modern Galactic dust maps constructed from different observational tracers and methodologies. We additionally develop a procedure for extending maps with incomplete sky coverage (for systematics estimation only) and use the low-redshift Hubble diagram as an empirical test of whether alternative extinction maps could be excluded by future large datasets.

A central result of this work is that the treatment of Milky Way extinction during SALT3 training is generally more important than, changing the extinction correction applied during light curve fitting. When the extinction map is changed only during fitting, much of the difference is absorbed by the fitted SN color parameter, with Δc closely tracking $-\Delta E(B - V)$ and differences only arising due to the differences in the color law of MW and SN dust. Consequently, the traditional approach of varying Milky Way extinction only at the light-curve fitting stage substantially underestimates the full effect of

5. CONCLUSION

Systematic	$\Delta w_{0,\text{sys}}$	$\sigma_{w_{0,\text{sys}}}$	$\sigma_{w_{0,\text{sys}}}/\sigma_{w_{0,\text{stat}}}$	$\Delta w_{a,\text{sys}}$	$\sigma_{w_{a,\text{sys}}}$	$\sigma_{w_{a,\text{sys}}}/\sigma_{w_{a,\text{stat}}}$
Stat + Sys	-0.014	0.010	0.705	0.088	0.025	0.272
Historical Method	0.009	0.008	0.571	-0.025	0.006	0.064

Table 5: Milky Way dust systematic contribution to the Year 10 cosmological fit, following the method of (Brout et al. 2022). For each parameter $\theta \in \{w_0, w_a\}$, $\Delta\theta_{\text{sys}}$ gives the shift in the best-fit value between the stat-only and stat+sys fits, $\sigma_{\theta,\text{sys}}$ gives the systematic contribution to the parameter uncertainty, and $\sigma_{\theta,\text{sys}}/\sigma_{\theta,\text{stat}}$ expresses that systematic contribution relative to the statistical uncertainty alone. *Stat + Sys* uses the systematic covariance matrix built from the six Milky Way extinction maps considered in this work (CSFD, Planck, DESI *gr* and *rz*, and Mudur 6 and 15) in addition to the nominal S11 map. *Historical Method* instead uses the covariance matrix from the widely used 5% $E(B - V)$ rescaling systematic, shown for comparison.

uncertainty in the MW dust corrections. A consistent systematic treatment therefore requires the same extinction assumptions to be propagated through both model training and fitting, while we find that propagating the dust-map uncertainty through the bias-correction simulations has negligible additional impact.

We further show that the large low-redshift SN sample expected from LSST provides a direct means of testing Galactic extinction maps. Because the low-redshift Hubble diagram is only weakly dependent on the assumed cosmological model, maps that introduce incorrect extinction corrections can be identified through the additional scatter they produce in SN distances. This provides an opportunity for future supernova samples not simply to propagate existing MW dust uncertainties, but to empirically constrain which MW-extinction maps are sufficiently accurate for precision cosmology.

The remaining uncertainty nevertheless has a substantial impact on dark energy constraints. For the full ten year LSST sample, including the diversity among the viable dust maps reduces the dark energy Figure of Merit by approximately 15%. By comparison, the historical prescription based on a 5% scaling of the nominal S11 during light-curve fitting only, reduces the Figure of Merit by only 4%. Thus, our more complete treatment produces an impact on the dark energy Figure of Merit more than three times larger than the conventional Milky Way extinction systematic.

While we have not explicitly recalculated all sources of systematic uncertainty in previous SN cosmology analyses, the magnitude of this increase provides useful context for current systematic error budgets. If the approximately threefold increase in the cosmological impact of Milky Way extinction found here were representative of the corresponding change in the DES analysis of Vincenzi et al. (2024), Milky Way extinction would have constituted the single largest source of systematic uncertainty in that analysis. Galactic extinction may therefore represent a substantially more important limitation

for precision SN cosmology than suggested by existing error budgets and the LSST-SRD.

These results also motivate a different strategy for constructing future SALT3 training samples. Simply increasing the size of the training sample will not necessarily reduce sensitivity to Milky Way extinction if the additional supernovae are distributed across sight lines with appreciable or poorly constrained Galactic reddening. Instead, future training samples should preferentially include SNe Ia located in regions of exceptionally low Milky Way extinction, where the dependence on the adopted Galactic dust map is intrinsically minimized. Such objects may be disproportionately valuable for training compared with otherwise similar SNe observed through larger dust columns. Dedicated target of opportunity follow up programs with LSST, Roman, DESI, 4MOST, or complementary facilities could be used to identify and obtain high quality spectrophotometric or multiband observations of these particularly-valuable events. In this sense, SNe Ia discovered along the cleanest Galactic sight lines should be regarded as especially important resources for future SN model training, rather than simply as one component of a larger statistical sample. Building a training set intentionally concentrated in these regions could reduce the imprint of Milky Way extinction assumptions on the trained model and enable more robust cosmological analyses with LSST and Roman. As supernova cosmology moves into the statistically-dominated regime enabled by LSST, careful systematic treatment and data collection will become even more essential.

6. ACKNOWLEDGEMENTS

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